Visual Analytics of the 1994 UCI Adult Census Dataset: Bayesian Imputation, Dimensionality Reduction and an Interactive Tableau Dashboard

Visual Analytics — Resit Coursework

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Abstract

We present an exploratory visual analysis of the 1994 UCI Adult census extract (48,842 records, 14 attributes) for a policy-analyst user, combining Bayesian multinomial imputation of missing employment attributes with two dimensionality-reduction projections (PCA and t-SNE) delivered as an interactive Tableau dashboard. The main finding is that reported *capital income*, not wage-related attributes, is the sharpest visual separator of high earners: the PCA projection splits the population into three strata by capital-gain/loss activity, and within the capital-gain stratum 61.4% of respondents earn over \$50K versus 18.9% in the no-capital-activity majority (base rate 23.9%). Within every stratum, income rises along a work/status composite (hours worked, education, age, sex). Occupation and education gradients are steep — from 3.7% (*Other-service*) to 47.6% (*Exec-managerial*) and from 15.9% (high-school) to 74.0% (professional school) — and the male high-income rate (30.4%) is nearly triple the female rate (10.9%).

1 Introduction

The Adult dataset, extracted from the 1994 US Census by Kohavi [5], records demographic and employment attributes together with a binarised income label (≤\$50K / >\$50K). It is a standard benchmark for classification, but classification accuracy alone tells an analyst little about *structure*: which population segments drive high income, how those segments overlap, and where the data itself is unreliable.

The intended user of this work is a **policy analyst** studying the socio-economic drivers of income inequality: a domain expert who is fluent in census vocabulary and comfortable with interactive dashboards, but who does not read code and should not have to trust opaque model outputs. Following Munzner's nested model [1, Ch. 4], this domain situation drives every downstream choice: the task abstraction (§3), the visual encoding idioms (§4), and the algorithms (PCA, t-SNE, NUTS).

Why visualization rather than summary tables? Anscombe's Quartet [1, pp. 8–9] is the canonical argument: datasets with identical summary statistics can differ radically in structure that is obvious the moment it is plotted. The census equivalent: a table of income-by-education means cannot reveal that capital-gain reporters form a separate, majority-high-income stratum of the population — the PCA projection makes this visible in a single view (§4).

The deliverable is a packaged Tableau workbook (`adult_dashboard.twbx`) with two projection views (`Proj_PCA`, `Proj_tSNE`), three group-comparison views with an absolute/proportion toggle (`Income_by_Education`, `Income_by_Occupation`, `Income_by_Age`), an imputation quality view (`Bayesian_Imputation_QA`), and an overview dashboard.

2 Data Preparation and Abstraction

The analysis dataset combines the UCI `adult.data` and `adult.test` files (48,842 rows, 14 features, one label). Combining them required stripping trailing periods from test-set labels and leading whitespace from all categorical values. The full pipeline is reproducible: staged Python scripts with printed verification gates, fixed random seeds, and all dimensionality-reduction hyperparameters logged to `params.json`.

Anomalies, quantified. The `?` sentinel marks missing values in exactly three columns: `workclass` (2,799), `occupation` (2,809) and `native-country` (857). The missingness is not homogeneous, and we name the two kinds separately. All 2,799 `workclass` gaps co-occur with `occupation` gaps (structural co-missingness of the employment block), and the 10 remaining `occupation` gaps belong to `Never-worked` respondents for whom no occupation exists — these are set directly to an explicit *None (never worked)* category rather than being modelled. Only `native-country` exhibits ordinary item missingness; its 857 gaps are recoded to an explicit *Unknown* category because a 41-class Bayesian model is disproportionate for a column that is not a projection input — a stated trade-off, not a hidden one. There are 52 full-row duplicates, retained as plausible identical census records. `education` and `education-num` are a redundant 1:1 pair; the ordinal `education-num` is kept for modelling and the text form retained only as tooltip context (a *derive* decision in Munzner's sense [1, pp. 49–53]).

Data abstraction. In Munzner's Ch. 2 vocabulary the dataset is a flat table of 48,842 items. Quantitative attributes: `age`, `fnlwgt`, `capital-gain`, `capital-loss`, `hours-per-week`; ordinal: `education-num`; categorical: `workclass`, `occupation`, `marital-status`, `relationship`, `race`, `sex`, `native-country`; the label `income` is *binary* categorical, which matters for the colour encoding (§4). Categorical/ordinal attributes act as keys and quantitative attributes as values — exactly the Dimensions/Measures split Tableau inherits from Polaris [1, p. 40].

Bayesian imputation. The employment attributes were imputed with sequential Bayesian multinomial logistic regressions in PyMC (NUTS sampler via nutpie; 1,000 posterior draws, 500 tuning steps, 4 chains, fixed seed): first `workclass`, then `occupation` with the (now complete) `workclass` as an additional predictor, mirroring their structural dependency. Predictors — standardised `age`, `education-num`, `hours-per-week`, `sex`, and marital status collapsed to married / previously married / never married — were chosen as complete columns plausibly associated with employment type. Each model was fitted on a stratified subsample of ≈8,000 observed rows (minimum 15 rows per class) to keep full NUTS sampling tractable; predictions cover every missing row. Convergence was checked before proceeding: maximum $\hat{R} = 1.000$ (min bulk ESS 1,094) for the workclass model and $\hat{R} = 1.010$ (min bulk ESS 777) for the occupation model, both within the pre-registered $\hat{R} \leq 1.05$ gate.

The standard "posterior mean / posterior std" imputation recipe assumes a continuous target; for these categorical targets we store the posterior-predictive *modal category* as the imputed value, and the modal probability plus predictive entropy as per-row uncertainty columns. This makes the imputation's confidence itself visualisable: mean modal probability was 0.756 for workclass but only 0.297 for occupation (14 classes), i.e. the model is honest that occupation is hard to recover. A known artefact of modal point imputation is surfaced rather than hidden: it over-assigns majority classes (99.8% of imputed workclass rows became *Private* versus 73.6% in the observed marginal). The `Bayesian_Imputation_QA` sheet plots imputed against observed distributions precisely so the analyst sees this shrinkage; the stored entropy columns quantify it per row. A mode-imputation baseline (which would place *100%* of rows in the majority class with no uncertainty estimate) is retained in the dataset as a clearly labelled fallback comparison only.

Projection feature space. Both projections use the same six standardised features — `age`, `education-num`, `hours-per-week`, log(1+`capital-gain`), log(1+`capital-loss`), `sex` — recorded verbatim in a `variables_used` column that every projection tooltip displays. Exclusions are deliberate: `income` is the conjectured categorisation the projections are meant to *verify* [1, §13.4.3] and would leak the answer into the layout; `fnlwgt` is a survey design weight describing the sampling frame, not the respondent; the high-cardinality nominals are surfaced through tooltips instead of being one-hot inflated into the distance metric.

3 Task Definition

Tasks are phrased in Munzner's action/target vocabulary [1, Ch. 3] so that each view can be checked against the question it exists to answer.

1. **Discover and verify cluster structure.** *Discover* whether the six-attribute feature space contains clusters or strata, and *verify* whether they align with the income category (targets: clusters, similarity). Answered by `Proj_PCA` and `Proj_tSNE`: the analyst inspects colour-coded structure, then uses tooltips to *identify* what individual points are.

2. **Compare income distributions across groups.** *Compare* the income distribution (target: distribution) across education levels, occupations and 5-year age bands — in absolute counts when the analyst asks "where are the people?" and proportions when asking "which group has the highest rate?". Answered by the three `Income_by_*` sheets with the Show As parameter toggle.

3. **Summarise imputation quality.** *Summarise* the derived (imputed) data and *compare* imputed against observed category distributions, so the analyst knows how much trust to place in the 5.7% of rows whose employment attributes are model output. Answered by `Bayesian_Imputation_QA`.

4 Visualisation Justification

We justify each view at the idiom level of the nested model [1, Ch. 4], citing the perceptual evidence for every channel choice.

Projections as scatterplots. Dimensionality reduction is attribute aggregation in Munzner's taxonomy [1, §13.4.3], and its display follows the document-collection pattern given there: high-dimensional table → 2D layout → scatterplot colour-coded by a conjectured categorisation, with the task of verifying whether visual clusters match it. Two-dimensional scatterplots are empirically the safest idiom for inspecting reduced data [2]; a 3D scatterplot was rejected on both this DR-specific evidence and the general argument against unjustified 3D for abstract data (depth is read far less accurately than planar position; occlusion and perspective distortion break the position and size channels) [1, Ch. 6.3].

Position on a common scale is the most accurate magnitude channel, so it is reserved for the projection coordinates themselves; `income`, a binary categorical attribute, takes hue — the highest-ranked identity channel after spatial position [1, Fig. 5.6]. Encoding a categorical attribute with an ordered channel (or vice versa) would violate the expressiveness principle [1, p. 100]. With 48,842 marks the scatterplot is beyond the "dozens to hundreds" comfort zone Munzner notes for scatterplots [1, §7.4]; 30% mark opacity turns overlap into readable density, which is what rescues the idiom at this scale.

Both projection layouts carry two caveats that the dashboard states and the analyst must internalise: DR layouts are affine invariant, so only *relative* distances carry meaning, never absolute position or orientation; and fine-grained distances are less reliable than large-scale cluster

Idiom	Proj_PCA / Proj_tSNE
What: data	Table, 48,842 items $\times$ 6 quantitative attributes
What: derived	2 synthetic attributes (PC1/PC2 or TSNE1/TSNE2); variables_used records the contributing attributes
How: encode	Point marks; aligned position for the 2 synthetic attributes; hue for binary income; 30% opacity against overplotting
How: facet	Juxtaposed in the overview dashboard; same data, same colour encoding, independent navigation
How: reduce	None (all items shown; aggregation happens in attribute space)
Why: task	Discover/verify income-aligned cluster structure; identify points via details-on-demand tooltips
Scale	48,842 marks

Table 1: Idiom analysis of the projection sheets, following the summary-table format Munzner uses for the equivalent DR scenario [1, p. 319].

structure because information is necessarily lost in the reduction [1, pp. 319–320]. The t-SNE view additionally shows thread-like micro-clusters: census attributes are quasi-discrete, so many records are exact duplicates in feature space — an artefact of the data's granularity, not evidence of fine social structure. t-SNE hyperparameters (perplexity 30, 1,000 iterations, PCA initialisation, fixed seed [4]) are logged in `params.json`.

Tooltips. Every projection mark carries a details-on-demand tooltip (age, education, occupation, income, hours/week, sex, and the `variables_used` string), implementing the overview-first, details-on-demand mantra [3][1, §6.7]. Displaying `variables_used` in the tooltip operationalises the requirement that DR plots communicate which original variables produced them — the synthetic axes are otherwise uninterpretable to the analyst.

Group comparison bars. The `Income_by_*` sheets are stacked bar charts: one categorical key (region: separated and ordered along an axis) and one quantitative value (aligned position), with hue for the income split [1, §7.5]. Ordering is a deliberate, named decision in each case: education is ordered by `education-num` (its inherent ordinal order), occupations are value-sorted to reveal the gradient, and age bands follow their natural sequence. The mandatory absolute / proportion toggle is one Show As parameter driving a calculated field (not two parallel sheets), and its justification is perceptual: absolute counts and proportions support different comparison tasks when group sizes differ [1, §5.6] — *Doctorate* holders are a tiny bar in absolute view yet the second-highest rate (72.6%) in proportional view.

Colour. The income palette (blue ≤$50K, orange >$50K) was validated computationally, not by eye: simulated protanopia/deuteranopia/tritanopia colour distances all exceed $\Delta E = 96$ and both colours clear 3:1 contrast on the workbook background. This matters because red–green colour-vision deficiency affects roughly 8% of males [1, p. 235], and the two hues also differ in luminance, satisfying the "get it right in black and white" rule [1, p. 140]. The palette ships as `Preferences.tps`. The QA sheet uses neutral grey for observed rows so the imputed (blue) distribution is the visually dominant comparison.

Imputation QA view. `Bayesian_Imputation_QA` juxtaposes the workclass distribution of imputed rows against observed rows as aligned bars sharing one axis — "eyes beat memory": simultaneous views beat remembered comparison [1, §6.5]. This view exists because imputation is a *derive* operation whose output the analyst must be able to audit; it is where the modal-imputation majority-shrinkage (§2) is visible instead of buried in a methods paragraph.

Dashboard composition. The overview dashboard is a juxtaposed, coordinated multi-view design [1, Ch. 12]: all views share the same underlying data and the same income colour encoding (shared encoding, all-data), while navigation is independent per view. The parameter control is placed once, above the views it drives. No 3D, gradients, shadows or decorative effects are used — function first, form next [1, §6.10].

5 Conclusion

What the analyst learns. The projections answer Task 1 affirmatively and specifically: the feature space is stratified, and the strata are economic. PCA's second component is dominated by capital income (+0.70 capital-loss, −0.63 capital-gain loadings), splitting the population into a capital-gain stratum (8.3% of respondents, 61.4% of whom earn >$50K), a capital-loss stratum (50.1% high earners) and the no-capital-activity majority (18.9%). Reporting *any* capital activity is thus a stronger high-income signal than any demographic attribute — a directly policy-relevant asymmetry between capital and wage income. Within every stratum, income rises along a work/status composite (PC1: hours worked 0.56, sex 0.47, education 0.39, age 0.36). The comparison views quantify the gradients (Task 2): by occupation, from 3.7% (*Other-service*) to 47.6% (*Exec-managerial*); by education, monotonically from 15.9% (high-school) through 41.3% (Bachelors) to 74.0% (professional school); by age, a life-cycle hump peaking at 40.2% for ages 50–54; by sex, 30.4% of men versus 10.9% of women — visible in the projections as the male-coded region of PC1. The QA view (Task 3) tells the analyst to treat imputed employment rows as low-confidence: occupation imputation carries mean modal probability 0.297, and imputed rows over-represent majority classes.

What we learned about large-scale visualization. Three lessons stand out. First, at ∼50K items every idiom decision becomes a scalability decision: raw scatterplots saturate, and opacity, aggregation or projection — not a bigger canvas — are what restore readability. Second, derived data needs its own visualization: both the DR coordinates and the imputed values are model outputs, and the coursework's most honest views (`variables_used` tooltips, the QA sheet, the entropy columns) exist to keep those derivations auditable rather than authoritative; visual analytics that hides its derivations is indistinguishable from decoration. Third, the perceptual rules are checkable: palette safety was *computed* (CVD simulation distances), convergence was gated on $\hat{R}$, and each pipeline stage printed a verification before the next ran — treating design guidance as testable constraints rather than taste made the workbook right on first open.

References

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